ÉCOLE UNIVERSITAIRE DE PHYSIQUE ET D'INGÉNIERIE
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M2PFA UP & NP
IMMAP
DU Data Scientist

Statistics Projects

Two-dimensional Normal distribution : measuring the correlation.

Let's consider a pair of correlated random variables x and y, with known expected values and variances : $\mathbb{E}[x] = \mu_x = \mathbb{E}[y] = \mu_y = 0$ and $\mathbb{V}[x] = \sigma_x^2 = 1$, $\mathbb{V}[y] = \sigma_y^2 = 4$. Both variables are distributed according the Normal law with an unknwon correlation, ρ, that we aim at determining. We have a sample of $N = 5 \times 10^3$ measurements, $\vec{r}_i = (x_i, y_i)$, $(i = 1, N)$, generated by the following python function :

```
# sample generator
def data(N=5000) :
    np.random.seed(154)      # for reproducibility
    M=[0.,0.]                # known expected values : E[x] = E[y] = 0
    Vx,Vy=1.0,4.0            # known variances       : Var(x)=1 ; Var(y)=4
    r=np.random.uniform(0,1) # 'unknown' correlation (KEEP IT BLIND - DO NOT PRINTOUT)
    Cxy=r * math.sqrt(Vx*Vy) # Covariance(x,y)
    COV = [ [ Vx , Cxy ], [Cxy, Vy ] ] # Covariance matrix
    XY = np.random.multivariate_normal(M, COV, size=N) # data generator  XY = {(x_i,y_i)}
    return XY,N
```

A blind analysis will be conducted, i.e. with no a-priori knowledge of the true correlation value that is (pseudo-randomly) chosen by the generator. The statistical model of the distribution is given by the bi-dimensional Normal law :

$$\mathcal{N}_{2D}(\vec{r};\rho) = \frac{1}{2\pi\sqrt{\det(C_{xy})}} e^{-\frac{\chi^2}{2}}$$

where C_{xy} is the covariance matrix, that depends on the unknown parameter ρ. The χ^2 variable is defined as $\chi^2 = (\vec{r} - \vec{\mu})^T C_{xy}^{-1} (\vec{r} - \vec{\mu})$ with $\vec{r} = (x, y)$ and $\vec{\mu} = (\mu_x, \mu_y)$ [1].

1. Derive an expression for the negative log-likelihood, $-\log\mathcal{L}(\rho)$, and implement it in the python code.
2. Using the minimization tools of the *scipy* module, determine the value $\hat{\rho}$ that minimizes $-\log\mathcal{L}(\rho)$, as well as the corresponding minimal value : $(-\log\mathcal{L})_{\min} = -\log\mathcal{L}(\hat{\rho})$.
3. We aim at evaluating the statistical uncertainty $\sigma_{\hat{\rho}}$ associated to the measured parameter $\hat{\rho}$. Draw the function curve $\Delta\log\mathcal{L}(\rho) = \log\mathcal{L}(\hat{\rho}) - \log\mathcal{L}(\rho)$ as a function of the parameter ρ and determine the two values $\rho_\pm$ that correspond to $\Delta\log\mathcal{L}(\rho_\pm) = \frac{1}{2}$, by scanning the curve in the vicinity of the minimum.
   Justify that we can define the uncertainty as $\sigma_{\hat{\rho}} = |\rho_+ - \rho_-|/2$.

1. If you prefer, you can also look in the litterature for an explicit definition of $\mathcal{N}_{2D}(\vec{r};\rho)$ as a function of the correlation parameter.

4. Produce a bi-dimensional representation of the sample measurements, (x_i, y_i). Draw on the same figure the ellipse of constant-χ^2 that corresponds to $\chi^2 = \alpha_{90} = \mathrm{Q}_{\chi^2}(\mathrm{p} = 90\%, \mathrm{n_{dof}} = 2)$, where $Q_{\chi^2}(p, n_{\mathrm{dof}})$ is the invert cumulative function (i.e. the percent-point function) of the χ^2 law with n_{dof} degrees of freedom. The orientation of the ellipse axis will be calculated using the estimated value $\hat{\rho}$.

5. **Coverage study** : we aim at evaluating, using a frequentist approach, whether the ellipse defined above correctly represents a 90% confidence level (CL) contour, meaning a contour that contains 90% of the random variable occurences when N tends to infinity. For that purpose, we count the number N_{90} of measurements (x_i, y_i) located inside the $\chi^2 = \alpha_{90}$ ellipse area among the N measurements of the sample [2]. Compute the ratio $\hat{p}_{90} = N_{90}/N$ and evaluate the associated statistical error, $(\sigma_{\hat{p}_{90}})_{\mathrm{stat.}}$, using the binomial law.

6. Evaluate the systematical error $(\delta_{\hat{p}_{90}})_{\mathrm{syst.}}$ that is induced by the uncertainty $\sigma_{\hat{\rho}}$ affecting the correlation parameter $\hat{\rho}$. We can choose, for instance, to repeat the above process to determine the variations of the ratio $\delta\hat{p}_{90}$ when the correlation value $\hat{\rho}$ (and thus the ellipse axis orientation) is varied by $\pm\sigma_{\hat{\rho}}$. Is the systematical error negligible with respect to the statistical error ?
Compare the final result, $\hat{p}_{90} \pm (\sigma_{\hat{p}_{90}})_{\mathrm{stat.}} \pm (\delta_{\hat{p}_{90}})_{\mathrm{syst.}}$, with the expected contour p-value : $\mathrm{CL} = p = 90\%$.

7. Repeat the determination of $\hat{p}_{\mathrm{CL}}$ and its associated statistical errror $(\sigma_{\hat{p}_{cl}})_{\mathrm{stat.}}$ for different ellipses with CL contours varying from 0.5% to 99.5% by steps of 3% (systematical uncertainty will be neglected at this stage). Plot the differences $\Delta p = \hat{p}_{\mathrm{CL}} - \mathrm{CL}$ with error-bars as a function of the expected CL. Conclusion.

8. **Unblinding** : You can now printout the true correlation value ρ^{true} as used at the generator level, and compare it to your measured value $\hat{\rho}$.
Subsidiary question : did you manage to conduct the whole analysis without any cursory indicative glance at the true correlation value ?

2. A possible method consists in applying the linear transfomation $\mathbf{U}$ that rotates into the eigenvectors basis of the correlation matrix C_{xy} and to count the number of points $\vec{r'}_i = \mathbf{U}\vec{r}_i$ satisfying

$$\frac{x'^2_i}{v_{x'}} + \frac{y'^2_i}{v_{y'}} \leq \alpha_{90}$$

where $v_{x'} = \sigma^2_{x'}$ and $v_{y'} = \sigma^2_{y'}$ are the C_{xy} eigenvalues.